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## **Project Title:**

Net Analytics: An Interactive NCAA Division 1 Basketball Analytics Dashboard

## **Project Summary**

Net Analytics is an NCAA basketball analytics dashboard that allows users to discover, compare, and analyze team/player performances across seasons. Net Analytics is a web-based application. It is based on historical and current NCAA Division 1 basketball data (from the Kaggle dataset) to provide structured access to statistics. For example, MVP, win/loss records, tournament outcomes, etc. The problem we solve is the lack of a unified, user-friendly platform that integrates advanced analytics into a single interactive application. Net Analytics will help users to access, compare, and visualize NCAA data in a friendly way.

## **Creative Component**

Interactive Comparative Analytics Visualizer and Prediction Tool.

We will implement:

- Query builder interface that allows users to use custom filters
- Predictive win probability model using previous data
- Dynamic multi-team analytics with interactive visualization

In terms of complexity, it requires complex SQL queries and the use of frameworks/tools for data analysis. The backend (we are planning to use Python for now) will turn our raw dataset into valuable analysis (a smart transformation).

## **Usefulness**

Our project will provide a centralized platform for exploring/analyzing NCAA Division 1 basketball data across teams, players, and seasons. Currently, fans, students, analysts, and sports journalists must rely on statistical pages or sports sites that just have spreadsheets of performance metrics. They lack advanced analytics and interactive tools that give meaningful comparisons and data.

Our application will allow users to search, filter, and compare teams and players using metrics such as win/loss records, ratings, tournament outcomes, scoring statistics, and more advanced metrics. Users will be able to visualize trends across seasons, and identify good players based on their strengths. Simple features include searching for a team or player and viewing season statistics, while more complex features include side by side comparings, trend graphs/visualizations, and ranking models based on different metrics.

There are existing sports analytic platforms out there, but those platforms sometimes lack some features our platform will have such as customizable comparison or interactive dashboard experience. Furthermore sometimes advanced analytics are restrictive to just teams/organizations and also behind paywalls too. Our platform will be accessible and open to all.

## Realness

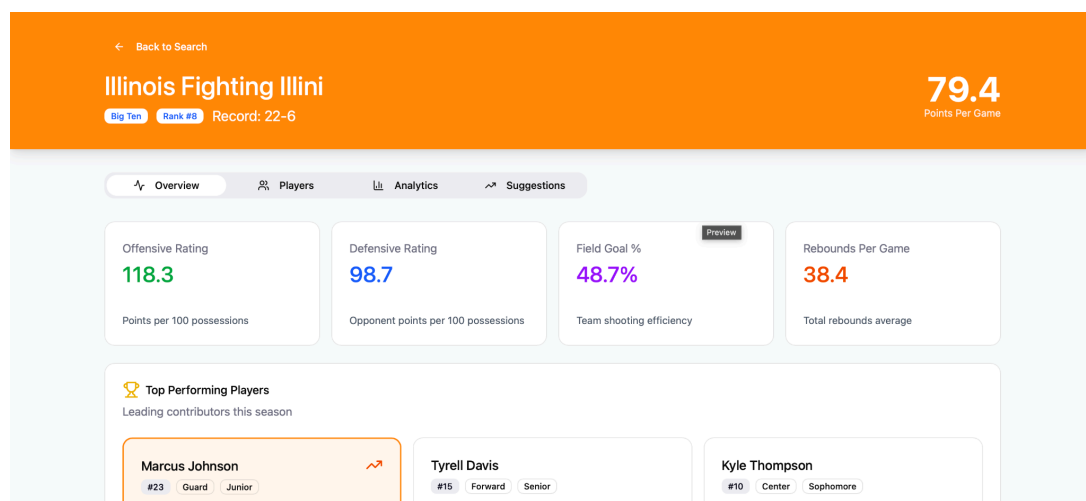
Our platform will be built using publicly available basketball datasets. Our main data set is the NCAA Basketball Dataset from Kaggle. This data is in the format of CSV files and the size of the data set is very large. There are more than 50,000 records of games , more than 350 records of teams and more than 10,000 records on players and their stats. The attributes include team names, conferences, seasons, game scores, outcomes, player points, rebounds , assists, tournament results, and seeding.

Our next data source will be from the NCAA official site. The format is also CSV files. This data set will be smaller than our primary data set. This dataset has more than 350 records and the attributes include conference affiliation, home arenas, and geographic locations.

## Functionality

The user (all users, regardless of affiliation with college basketball, or NCAA, could be fans, students, etc) should be able to open the web app by visiting the domain it is hosted on. Then, a user will be faced with a splash screen, prompting the user to either search for a team to analyze (Read), or can view top performing teams this season and click to view why (i.e. team defensiveness, etc.) For each team, they can view top performing players, top team stats, and which players are responsible for those team stats. In addition, there should be a section for smart suggestions, focusing on which aspects of basketball the team should focus on, or individual player suggestions based on past games. NCAA Administrators should be able to import data (Create) for all teams, and make corrections if necessary (Update). In situations where teams wins/losses are vacated due to extreme circumstances, NCAA administrators should be able to remove (Delete) data from those seasons.

For the sake of visualization, a mockup has been created with Figma for the Fighting Illini Men's basketball team



For now, we hope to divide the work evenly, with one student working on the backend, another student working on the front end, and together, we plan on diagraming the relations of all tables, implementing the database, and determining an ORM framework to link the backend to the database. We will brainstorm together to determine how to overcome the technical challenges of implementing the smart suggestions feature.